homework 7

Spring 2025-AZ10

5 Provide a brief answer to each

[1] Define: Supervised machine learning

- Supervised machine learning is a type of machine learning where we use labeled training data and specific algorithms to find an already known solution. When the patterns are developed, these are used for prediction in news unseen Data.

[2] Define unsupervised machine learning

- unsupervised machine learning where unseen, unlabeled data without correct out comes is used to train an algorithm. This is to find new unseen patterns

[3] what is linear regression

- Linear Regression is a SUPERVISED Learning Technique that predicts relationships using a straight line on one or more inputs

[4] What is logistic Regression

- Logistic Regression is a Supervised learning technique that uses classification. It groups data into 0's and 1's

[5] What is K-means Clustering

- K-means clustering is an unsupervised learning technique that groups the data using "K" groups, where data points are grouped to the nearest "K" points. Ultimately, going until its Stabilized

[6] what is K-nearest neighbor

KNN is a Supervised machine learning algorithm that classifies new data points by how Similar it is to groups by examining K most similar points in training dataset

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5 [7] What is an artificial Neural Network?

- An ANN is a machine learning model that mimics the human brain. It connects artificial neurons to process data and adjust the weights to learn patterns

[8] Define basic statistics: average, variance, Standard Deviation

- Average is taking all of the values, adding them together, and then dividing the new value by how many values there originally was
- Variance Measures how Spread out values are by Calculating average Squared Distance from the mean
- Standard Deviation is the Square root of Variance

[9] What is a Support Vector Machine?

- A SVM is a Supervised machine learning algorithm that Classifies data by finding a decision boundary that maximizes the margin between Different Classes. The closest Data points, called Support Vectors, Determine that Boundary

[10] What is an regular expression? whats it good for?

- A regular Expression is a sequence of Characters that Defines a search pattern for Text. It is used to find, match, validate, or replace text that follows a specific pattern

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4 What Does the code print when you run it?

- First part: the "*" is greedy and finds the first "o" and last "o" and takes everything cutting off what was before the first "o" and after the last "o"

=> ["o be or not to be ought to not be the other questio"].

- Then the "*?" makes it lazy and takes the first "o" and finds the nearest "o" and repeats that for all "o"'s

=> ["o be o", "ot to", "ought to", "ot be the o"

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4 Second Part:

- BD
- AC

A310 Midterm Fall 2024

4 Write a program that would determine experimentally the probability:

1) The probability of getting 6 heads in a row when you throw a coin 100 Time?

```
Import random

def has_six_heads(flips):
    Count = 0

    for flip in flip:
        If flip == "H":
            Count += 1
        else:
            Count = 0

        if Count == 6:
            return True
    return False

def Simulate(trials = 10 000):
    Successes = 0

    for i in range(trials):
        flips = []
        for i in range(100)
        flips.append(random.choice(["H", "T"]))
        if has_six_heads(flips):
            Successes += 1
    return Successes/trials
print(Simulate())
```

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4 1) 54 %

Explain what you would have to change if you wanted to calculate instead:

(54%) a) Six tails in a row

- change the program to count "T" values instead of "H" values, the probability would be the same since its a fair coin

(28%) b) Six tails in a row and Six heads in a row

- modify to count both streaks and only be successful when both occur

(80%) c) Six tails in a row or Six heads in a row

- modify to count both streaks and be successful when one occurs

(26%) d) Six tails in a row but not Six heads in a row

- Count an experiment as successful if a six-tail streak occurs and no six head streak occurs in the same experiment

5 Implement a LinkedList ADT in Python. Include test and expected outcomes

```
Class Node:
    def __init__(self, value):
        self.value = value
        self.next = None

Class LinkedList:
    def __init__(self):
        self.head = None

#INSERT: add a node to the end
    def insert(self, value
        new_node = Node(value)
```

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5

```
#Insert
    def insert(self,value):
        new_node = Node(value)

        if self.head is None:
            self.head = new_node
            return

        current = self.head

        while current.next is not None:
            current = self.head

        current.next = new_node

#FIND: Search for value
    def find(self,value):
        current = self.head

        while current is not None:
            if current.value == value:
                return True

            current = current.next

        return False
```

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```
# Remove: Delete first matching value
    def remove(self, value)
        if self.head is None:
            return False

        if self.head.value == value:
            self.head = self.head.next
            return True

        current = self.head

        while current.next is not None:
            if current.next.value == value:
                current.next = current.next.next
                return True

            current = current.next

        return False

#Show: Display the LinkedList
    def show(self):
        values = []
        current = self.head

        while current is not None:
            values.append(str(current.value))
            current = current.next

        print("->".join(values + ["None"]))
```

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5 Test the program

Code:

```
LL = LinkedList()

# Test empty list
LL.Show

# insert values
LL.insert(10)
LL.insert(20)
LL.insert(30)
LL.Show

# Find existing And Missing values
print(LL.find(90))
print(LL.find(99))

# Remove Middle node:
print(LL.remove(90))
LL.Show()

# Remove head
print(LL.remove(10))
LL.Show

# remove last node
print(LL.remove(30))
LL.Show

# Try Remove from empty list
print(LL.remove(10))
```

Results:

```
None
10 -> 20 -> 30 -> None
True
False
True
10 -> 30 -> None
True
30 -> None
True
None
False
```

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LQ

```
def longest(seq):
    if not seq:
        return []

    result = [seq[0]]

    for i in range(1, len(seq)):
        if seq[i] >= seq[i-1]:
            result.append(seq[i])
        else:
            break

    return result

def rest(seq):
    return seq[len(longest(seq)):]

def merge(a,b):
    if not a:
        return b
    if not b:
        return a

    if a[0] <= b[0]:
        return [a[0]] + merge(a[1:], b)
    else:
        return [b[0]] + merge(a, b[1:])

def sort(seq):
    if not seq:
        return []

    return merge(longest(seq), sort(rest(seq)))
```

#Test → `print(sort([4,5,2,3,1,7,6]))`